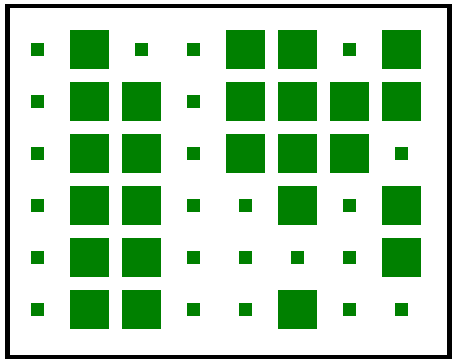


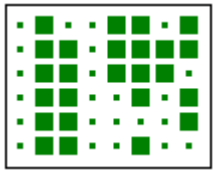


Lab 2
Make and Flex

CS 330
Programming Languages and Compilers

Team Lab Assignment

Obtain the Lab 2 Team Starting Code

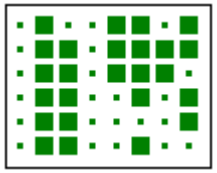


- Open a terminal and change directory into your **`cs330-TeamLabWork`** directory
- Use git to obtain the lab 2 team code:

```
git pull
```

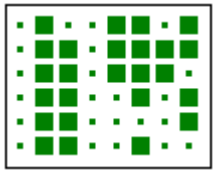
- The directory **`lab2-Flex-Team`** should be added to your project
  - It contains the initial calculator we worked on in class
- Your team is adding more operations to the calculator

Team Lab Assignment: t-1



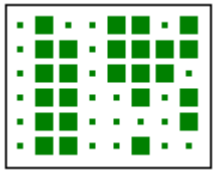
- Do this work in **`lab2-Flex-Team`**
- Extend the calculator
  - Note that it need *not* respect PEMDAS
- Each team member is to implement two unique mathematical functions in their own separate C file which are linked to the calculator
  - File names should be `calc_`*`studentid`*`.c`
    - i.e. `calc_jdoe.c`
    - header will be `calc_`*`studentid`*`.h`
  - Add comments to clarify what the functions do

Team Lab Assignment: t-1 (cont'd)



- The token and regex lists must be expanded to include syntax to call all of the functions – including the tokens already in the flex file
  - You will need to cooperate so updates to the flex file don't collide between team members
- When you compile the `lex.yy.c` file you'll need to link it with the separate C files containing the team members' functions
    - Remember that the flex input file will need to include function declarations for each of the external functions **(the .h files)**

Team Lab Assignment: t-1 (cont'd)

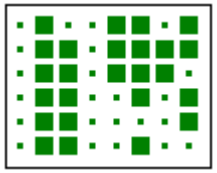


- To use the math library you must include the **`-lm`** argument to the C compiler
  - That is the letter "`l`" not a one (e.g. library math)
- For example, to compile the calculator with John Doe's additions:

```
flex calc_implementation.l
cc -c lex.yy.c
cc -c add_sub.c
cc -c calc_jdoe.c
cc -lm lex.yy.o add_sub.o calc_jdoe.o
```

Submitting the Team Lab Work

Submitting the Team Lab (1 of 2)



- The student who hosted the latest editing session, using the terminal, in the **`cs330-TeamLabWork`** directory, use the following commands:

```
git add .
```

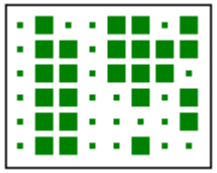
  - Note the trailing dot (preceded by a space) is needed

```
git commit -m "Updated lab work"
```

```
git push
```

  - The first time you push you may be asked to set the default branch. Enter the command as given in the message.

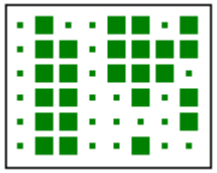
Submitting the Team Lab (2 of 2)



- After the code has been pushed (previous slide), the students who did not host the latest editing session should obtain the updated code, using the terminal, in the project directory, use the following command:
  `git pull`
- When the team has completed the lab, the team's understanding leader for this lab creates and pushes an annotated tag to indicate the work is completed (the assignment isn't submitted until this is done):
  `git tag -a Lab2Submission1 -m "Lab completed"`

  `git push --follow-tags`
- Team members should receive an email within an hour verifying the lab was received for grading

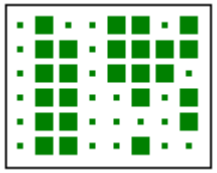
Errors When Using Git



- If an error occurs please capture a screenshot of the error and send that to the instructor along with a overview of what you were trying to do
  - Generally the errors are related to not being in the correct directory when using the Git commands

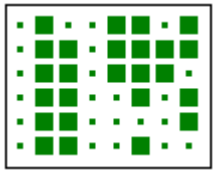
Individual Lab Assignment

Lab Submission



- For the problems on the following pages, each answer should be in its own file named `lab2prob-i-#.l`
  - Where # is the problem identifier
    - For instance, for individual problem 1 the file will be named **`lab2prob-i-1.l`**
- You submit your work via `git` as described after the lab steps

Individual Lab Assignment: i-1



- Do this work in **`lab2-Flex-Introduction`**
- Create a scanner which translates texting slang into English (you may use `printf` in your actions, they will not return tokens in that case)
- Support at least 10 texting terms, including “irl”
  - e.g. **`irl`** (or `Irl`, `IRL...`) translated to **`in real life`**
- Output any non-matching text intact
- Match entire “words” **(flex does not support \b)**
  - e.g. **`twirl`** should not become **`twin real life`**

Individual Lab Assignment: i-1 (cont'd)



- The resulting program should take input like:

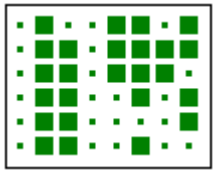
```
Idk if the falling and swirling
leaves are gr8 irl. Imo winter is
too soon!
```

- And output it as:

```
I don't know if the falling and
swirling leaves are great in real
life. In my opinion winter is too
soon!
```

Submitting Individual Lab Work

Submitting the Individual Lab (1 of 2)



- Using the terminal, in the **`cs330-IndividualLabWork`** directory, use the following commands:

```
git add .
```

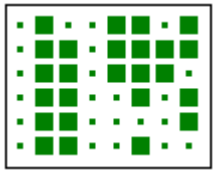
  - Note the trailing dot (preceded by a space) is needed

```
git commit -m "Updated lab work"
```

```
git push
```

  - The first time you push you may be asked to set the default branch. Enter the command as given in the message.

Submitting the Individual Lab (2 of 2)

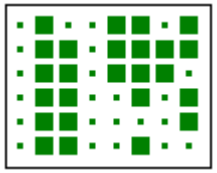


- When you have completed the lab, including following all steps on the previous page after your last changes are made, create and push an annotated tag to indicate the assignment is completed
- Assignment isn't submitted until this is done:

```
git tag -a Lab2Submission1 -m "Lab completed"
git push --follow-tags
```

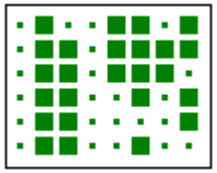
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Errors When Using Git

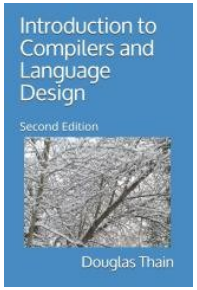


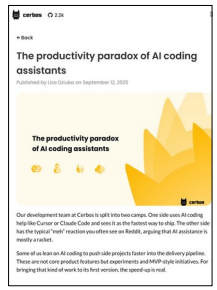
- If an error occurs please capture a screenshot of the error and send that to the instructor along with a overview of what you were trying to do
  - Often the errors are related to not being in the correct directory when using the Git commands

Homework



- Friday (10/2):
  - Compilers book Chapter 4
  - LLM: The productivity paradox…





- Sunday (10/4)
  - Journal entry for week 4
    - **2-1 Rule** (minimum 2 paragraphs, 1 quiz question)
- Monday (10/5):
  - Lab 2 Due: Flex & Make (Team and Individual)